

$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$B \quad C$   
 $2 \times 3 \quad 3 \times 1$   
 product  $2 \times 1$

$$= \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$$

$$= 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 \\ 0 \end{pmatrix}$$

$$A \quad B = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 0 \end{pmatrix} \perp \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 2 \end{pmatrix} & 0 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 0 \\ 2 \end{pmatrix} & 3 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 2 \end{pmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & -3 \\ 1 & 2 & 3 \end{pmatrix}$$

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$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

Can  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$  be written as a linear combination of  $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$  and  $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ ?

Does  $\begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$  have a solution?

Does the system of linear eqns  $\begin{cases} -x_1 + 0x_2 = 1 \\ x_1 + 2x_2 = 2 \end{cases}$  have a solution?

$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$DA = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} \boxed{-1} & \boxed{0} \\ \boxed{1} & \boxed{2} \end{pmatrix} = \begin{pmatrix} \boxed{1} & \boxed{4} \end{pmatrix}$$

$1 \times 2 \qquad \qquad 2 \times 2 \qquad \qquad 1 \times 2$

$$= 1 \begin{pmatrix} -1 & 0 \end{pmatrix} + 2 \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \end{pmatrix} + \begin{pmatrix} 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 4 \end{pmatrix}.$$

Solve

$$-x_1 + 0x_2 = 1$$

$$x_1 + 2x_2 = 2$$

$$\begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

$$F = \begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \quad \underline{z} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

1. Can  $\begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$  be written as a linear combination of  $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$  and  $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ ?

$\hat{=}$  Can I choose numbers  $x_1$  and  $x_2$  so that

$$x_1 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix}$$

$1 \times 2 \qquad \qquad 2 \times 2$

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$3 \times 2 \qquad 2 \times 1$

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$x_1 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$\begin{pmatrix} 2x_1 - x_2 \\ x_1 + x_2 \\ 0 + 2x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ x_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_2 \\ 2x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$\begin{cases} 2x_1 - x_2 = 3 \\ x_1 + x_2 = 0 \\ 0 + 2x_2 = -2 \end{cases}$$

$$x_1 = 1 \quad x_2 = -1.$$

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$$\begin{pmatrix} 2 & -1 & 3 & 3 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & -2 & -2 \end{pmatrix}$$